the basic vocabulary a processor uses to execute operations—it gives chip designers a common foundation they can build on and extend with custom features, without being locked into a licensing model controlled by a single company.

Bridging RISC-V CPUs and Nvidia GPUs

An AI server does not rely on its graphics processing unit (GPU) alone. The central processing unit (CPU) still manages tasks and coordinates with the GPU, which handles the bulk of the AI workload. As models grow larger, the speed at which these processors can exchange data becomes increasingly important.

Nvidia's NVLink Fusion is designed to help custom CPUs and accelerators communicate efficiently within high-performance computing systems. SiFive is working to bring this technology to its RISC-V-based solutions. Jensen Huang, founder and CEO of Nvidia, says, "This provides the flexibility to combine customisable RISC-V CPUs with Nvidia accelerators."

For customers, this enables a customised RISC-V CPU to work alongside Nvidia GPUs as part of the same AI computing system. Instead of treating the CPU and GPU as completely separate components, the two could be designed to work more closely together.

However, this integration won't happen immediately. Patrick told Reuters that its NVLink-enabled designs are not expected to reach the market before 2027.

Hardware is only half the story. Developers also need software that can actually put combined CPU-GPU systems to work. This is where Nvidia's CUDA platform becomes important. In 2025, the company announced that it was porting CUDA to RISC-V, using SiFive's HiFive Premier P550 as an early development platform.

CUDA provides the software tools and libraries developers use to build applications for Nvidia GPUs, so extending it to RISC-V could make it easier for developers to use a RISC-V CPU alongside an Nvidia GPU without having to build an entirely separate software environment.

As Patrick notes, "Without strong software support, the effort for companies to adopt new hardware technologies is simply too great."

If this software support eventually matures into production-ready systems, the combination of RISC-V CPUs, Nvidia GPUs, NVLink, and CUDA could give data-centre operators a more customisable way to build AI computing systems.

But Nvidia already has its own CPU—so why RISC-V?

This raises an important question. If Nvidia already ships its own Arm-based data-centre CPU platform, why bother with RISC-V at all?

The answer lies in customisation. Nvidia's Arm-based CPUs give data-centre customers a processor platform, while RISC-V could allow customers to develop CPUs more closely around their own requirements.

For hyperscalers running large data centres, this flexibility has real value. Instead of using the same general-purpose CPU design for every workload, a company could develop a processor with features suited to the way its systems operate.

This is where the SiFive-Nvidia collaboration becomes significant. A customer could potentially design a specialised RISC-V CPU, connect it to Nvidia GPUs through NVLink Fusion, and use Nvidia's software ecosystem for AI workloads.

In this model, Nvidia would not have to provide every processor in the system. It can let customers own the CPU design while Nvidia continues supplying the GPUs, the interconnect, and the software that ties accelerated computing together. That also means this isn't really RISC-V positioning itself as an Arm replacement. It's more about giving data-centre customers an additional lever to customise the CPU side of systems built around Nvidia's AI accelerators.

What comes next for Nvidia and SiFive?

The Nvidia-SiFive partnership is still at an early stage. NVLink-enabled designs are not expected to reach the market before 2027, and neither company has disclosed which processors or customers will be first in line.

Still, the direction is clear: RISC-V is pushing beyond the embedded and low-power markets where it's traditionally been strong, and into high-performance computing territory. For Nvidia, backing RISC-V widens the pool of custom CPU designs compatible with its AI platform. For SiFive, access to Nvidia's interconnect and software ecosystem opens a door into data-centre systems where performance, customisation, and AI acceleration increasingly go hand in hand.

The real question going forward isn't whether RISC-V can compete with established CPU architectures anymore. It's whether RISC-V can become the customisable CPU foundation for AI systems—while working alongside the GPUs and software platforms that already dominate accelerated computing

For Nvidia, supporting RISC-V would make its AI platform compatible with a wider range of customised CPU designs. For SiFive, access to Nvidia's interconnect and software ecosystem opens a door into data-centre systems where performance, customisation, and AI acceleration increasingly go hand in hand.

The bigger question is no longer simply whether RISC-V can compete with established CPU architectures. It is whether it can become the customisable CPU foundation for AI systems, while working alongside the GPUs and software platforms that already dominate accelerated computing. EFY

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